

BDA Quiz-21 2025

* Indicates required question

1. Email *

2. Roll Number *

3. Password *

Question

4. We wish to use the Flagolet-Martin algorithm to count the number of distinct elements in a stream. Suppose that there are ten possible elements, 1, 2,..., 10, that could appear in the stream, but only four of them have actually appeared. To make our estimate of the count of distinct elements, we hash each element to a 4-bit binary number. The element x is hashed to $3x + 7$ (modulo 11). For example, element 8 hashes to $3 \cdot 8 + 7 = 31$, which is 9 modulo 11 (i.e., the remainder of $31/11$ is 9). Thus, the 4-bit string for element 8 is 1001. 1 point

A set of four of the elements 1 through 10 could give an estimate that is exact (if the estimate is 4), or too high, or too low. You should figure out under what circumstances a set of four elements falls into each of those categories. Then, identify in the list below the set of four elements that gives the exactly correct estimate.

Check all that apply.

- ☐ { 1, 3, 9, 10}
- ☐ { 1, 6, 7, 10}
- ☐ {1, 5, 8, 9}
- ☐ {2, 3, 6, 9}

This content is neither created nor endorsed by Google.

Google Forms